

第四章

4.2 $Z = \overline{X_1 \cdot X_2 \cdot Y_1 \cdot \bar{Y}_3}$

$Y_1 = X_1 + Y_1$

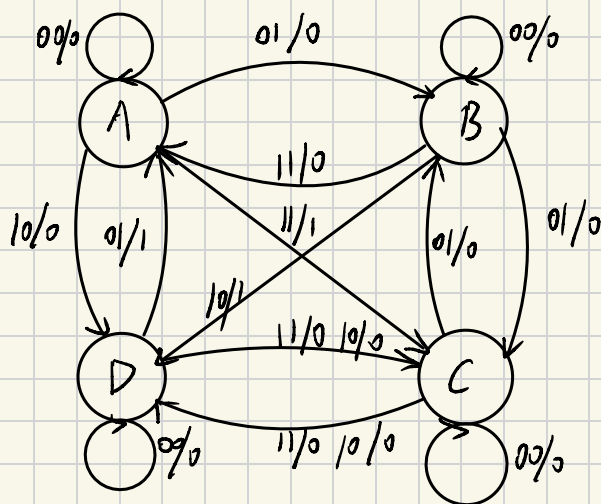
$Y_2 = X_1 \oplus Y_2$

$T = Z = \overline{X_1 \cdot X_2 \cdot Y_1 \cdot \bar{Y}_3}$

$K = Y_2 = X_1 \oplus Y_2$

$J = D = Y_1 = X_1 + Y_1$

4.3



4.6 输出函数

$Z_1 = Q_1$

$Z_2 = Q_2$

激励函数

$$J_1 = Q_1 \cdot \bar{Q}_2$$

$$K_1 = X \cdot Q_2 + Q_1 \cdot \bar{Q}_2$$

$$J_2 = \bar{X} \cdot Q_2$$

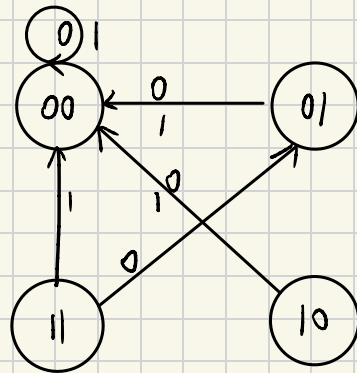
$$K_2 = Q_2$$

下面列出状态转移真值表

输入 X	现态 Q_2, Q_1	激励函数				次态	
		J_2	K_2	J_1	K_1	$Q_2^{(n+1)}$	$Q_1^{(n+1)}$
0	0 0	0	0	0	0	0	0
0	0 1	0	0	1	1	0	0
0	1 0	1	1	0	0	0	0
0	1 1	1	1	0	0	0	1
1	0 0	0	0	0	0	0	0
1	0 1	0	0	1	1	0	0
1	1 0	0	1	0	1	0	0
1	1 1	0	1	0	1	0	0

下面给出状态表及状态图

现态 Q_2, Q_1	次态 $Q_2^{(n+1)}, Q_1^{(n+1)}$	
	$X=0$	$X=1$
0 0	0 0	0 0
0 1	0 0	0 0
1 0	0 0	0 0
1 1	0 1	0 0



由状态图和状态表可知

该同步时序电路的功能为当且仅当状态为 11 输入为 0 时输出

为0 其他状态及输入均为置0操作

4/1

B	X						
C	X	X					
D	BE	X	X				
E	X	AD CF	X	X			
F	X	X	AD	X	X		
G	X	X	X	X	X	X	
H	X	X	X	X	X	X	GH BC
	A	B	C	D	E	F	G

由隐变量可知

等效状态有 (A, D), (B, E), (C, F), (G, H)

现态	次态/输出	
	x=0	x=1
A	B/0	A/0
B	A/1	C/0
C	C/0	A/1
G	H/1	G/1
H	C/1	B/1

4/13

y \ x	0	1
0	A	D
1	B	C

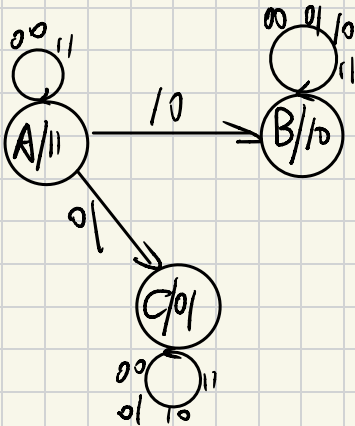
B C

A B C

B D

状态	次态/输出	
	x=0	x=1
00	00/0	01/0
01	10/0	01/0
11	10/0	01/0
10	01/1	00/0

4/15 状态图



原始状态表

现态	次态/输出		
	$X_1=0, X_2=0, 1$	$X_1=1, X_2=0$	$X_1=0, X_2=1$
A	A/11	B/10	C/01
B	B/10	B/10	B/10
C	C/01	C/01	C/01

状态编码

	0	1
0		B
1	C	A

则A的编码为11 B的编码为10 C的编码为01

则编码后的状态表为

现态	次态/输出		
	$X_1=0, X_2=0, 1$	$X_1=1, X_2=0$	$X_1=0, X_2=1$
11	11/11	10/10	01/01
10	10/10	10/10	10/10
01	01/01	01/01	01/01

采用D触发器

激励函数和输出函数真值表

$X_1 X_2$		00				01				11				10			
$y_2 y_1$		$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$	$y_2 y_1$	$D_2 D_1$
00			d d		d d		d d		d d		d d		d d		d d		d d
01		0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1	0 1
10		1 0	1 1	1 0	1 0	1 0	1 0	1 1	1 0	1 1	1 0	1 1	1 0	1 1	1 0	1 1	1 0
11		1 1	1 0	1 1	1 0	1 0	1 1	1 0	1 1	1 0	1 1	1 0	1 1	1 0	1 1	1 0	1 1

$X_1 X_2$		D_2				D_1			
$y_2 y_1$		00	01	11	10	00	01	11	10
00		d	d	d	d	d	d	d	d
01		0	0	0	0	1	1	1	1
11		1	0	1	1	0	0	1	0
10		1	1	1	1	0	0	0	0

化简可得

$$D_2 = \bar{y}_1 + x_1 y_2 + \bar{x}_2 y_2$$

$$D_1 = \bar{y}_2 + \bar{x}_1 y_1 + x_2 y_1$$

